使用 NetApp 檔案物件雙重性和 AWS SageMaker 進行雲端資料管理

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TR-4967 ：使用 NetApp 檔案物件雙重性和 AWS SageMaker 進行雲端資料管理

NetApp的Karthithkeyan Nagalingam

資料科學家和工程師通常需要存取 NFS 格式中儲存的資料、但直接從 AWS SageMaker 中的 S3 傳輸協定存取這些資料可能很困難、因為 AWS 只支援 S3 儲存區存取。不過、NetApp ONTAP 可為 NFS 和 S3 啟用雙傳輸協定存取、提供解決方案。有了這個解決方案、資料科學家和工程師就能透過 NetApp Cloud Volumes ONTAP 的 S3 儲存區、從 AWS SageMaker 筆記型電腦存取 NFS 資料。這種方法可讓您輕鬆從 NFS 和 S3 存取和共用相同的資料、而不需要額外的軟體。

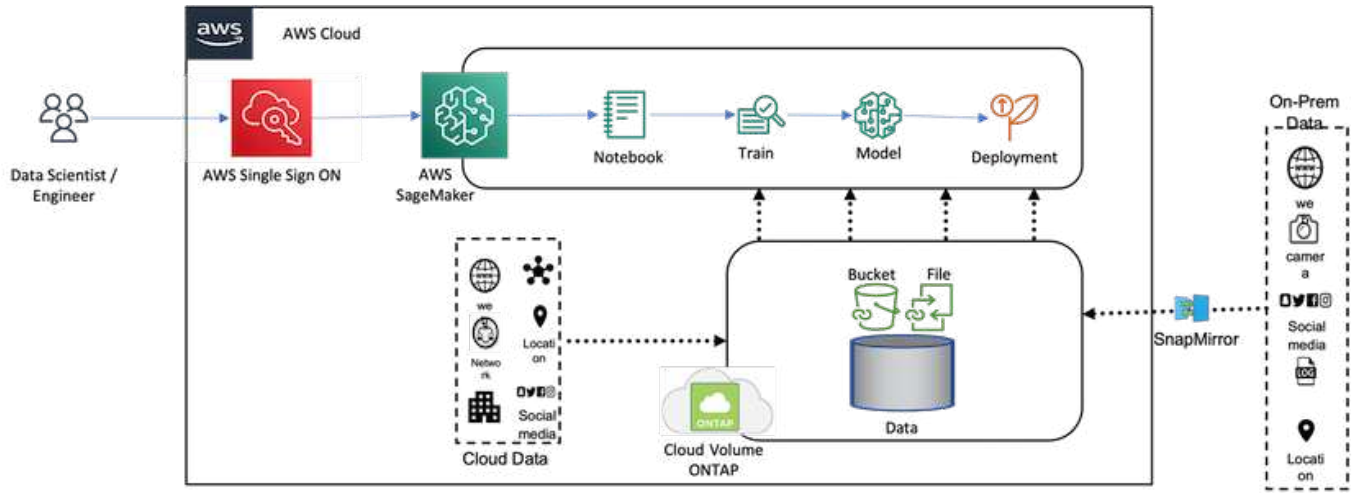
解決方案技術

本解決方案採用下列技術：

- * AWS SageMaker Notebook 。 * 為開發人員和資料科學家提供機器學習能力、以有效建立、訓練和部署高品質的 ML 模型。
- * NetApp BlueXP. * 可在內部部署、 AWS 、 Azure 和 Google Cloud 上探索、部署及操作儲存設備。它可提供資料保護、防範資料遺失、網路威脅和非計畫性中斷、並最佳化資料儲存和基礎架構。
- * NetApp Cloud Volumes ONTAP 。 * 在 AWS 、 Azure 和 Google Cloud 上提供企業級儲存磁碟區的 NFS 、 SMB/CIFS 、 iSCSI 和 S3 傳輸協定、讓使用者能夠更靈活地存取和管理雲端中的資料。

NetApp Cloud Volumes ONTAP 是從 BlueXP 建立來儲存 ML 資料。

下圖顯示解決方案的技術元件。



使用案例摘要

NFS 和 S3 的雙傳輸協定存取可能的使用案例、是在機器學習和資料科學領域。例如、資料科學家團隊可能正在使用 AWS SageMaker 進行機器學習專案、這需要存取 NFS 格式儲存的資料。不過、資料也可能需要透過 S3 儲存區來存取和共享、以便與其他團隊成員協同作業、或與其他使用 S3 的應用程式整合。

藉由使用 NetApp Cloud Volumes ONTAP 、團隊可以將資料儲存在單一位置、並可透過 NFS 和 S3 傳輸協定進行存取。資料科學家可直接從 AWS SageMaker 存取 NFS 格式的資料、而其他團隊成員或應用程式則可透過 S3 儲存區存取相同的資料。

這種方法可讓您輕鬆且有效率地存取和共用資料、而無需在不同的儲存解決方案之間進行額外的軟體或資料移轉。它也能讓團隊成員之間的工作流程和協同作業更有效率、進而更快、更有效地開發機器學習模式。

資料科學家和其他應用程式的資料雙重性

資料可在 NFS 中取得、並可從 AWS SageMaker 從 S3 存取。

技術需求

您需要 NetApp BlueXP 、 NetApp Cloud Volumes ONTAP 和 AWS SageMaker 筆記型電腦來處理資料雙重用途使用案例。

軟體需求

下表列出實作使用案例所需的軟體元件。

軟體	數量
藍圖	1.
NetApp Cloud Volumes ONTAP	1.
AWS SageMaker 筆記型電腦	1.

部署程序

部署資料雙重性解決方案涉及下列工作：

- BlueXP Connector
- NetApp Cloud Volumes ONTAP
- 用於機器學習的資料
- AWS SageMaker
- 通過 Jupyter 筆記型電腦驗證的機器學習

BlueXP 連接器

在此驗證中、我們使用 AWS 。也適用於 Azure 和 Google Cloud 。若要在 AWS 中建立 BlueXP Connector 、請完成下列步驟：

1. 我們使用的認證是以 BlueXP 中的 mcarl-Marketer-訂閱 為基礎。

2. 選擇適合您環境的區域（例如、 us-east-1 [N.）、然後選擇驗證方法（例如、承擔角色或 AWS 金鑰）。在此驗證中、我們使用 AWS 金鑰。
3. 提供連接器的名稱並建立角色。
4. 根據您是否需要公有 IP 、提供 VPC 、子網路或金鑰組等網路詳細資料。
5. 提供安全性群組的詳細資料、例如從來源類型存取 HTTP 、 HTTPS 或 SSH 、例如 Anywhere 和 IP 範圍資訊。
6. 檢閱並建立 BlueXP Connector 。
7. 確認 BlueXP EC2 執行個體狀態在 AWS 主控台中執行、然後從 * 網路 * 索引標籤檢查 IP 位址。
8. 從 BlueXP 入口網站登入 Connector 使用者介面、或使用 IP 位址從瀏覽器存取。

NetApp Cloud Volumes ONTAP

若要在 BlueXP 中建立 Cloud Volumes ONTAP 執行個體、請完成下列步驟：

1. 建立新的工作環境、選取雲端供應商、然後選取 Cloud Volumes ONTAP 執行個體類型（例如單一 CVO 、 HA 或 Amazon FSxN for ONTAP ）。
2. 提供 Cloud Volumes ONTAP 叢集名稱和認證等詳細資料。在此驗證中、我們建立了一個名為的 Cloud Volumes ONTAP 執行個體 `svm_sagemaker_cvo_sn1`。
3. 選取 Cloud Volumes ONTAP 所需的服務。在此驗證中、我們選擇僅監控、因此我們停用了 * 資料感知與法規遵循 * 和 * 備份至雲端服務 * 。
4. 在 * 位置與連線 * 區段中、選取 AWS 區域、 VPC 、子網路、安全性群組、 SSH 驗證方法、 以及密碼或金鑰配對。
5. 選擇充電方式。我們使用 * Professional* 進行此驗證。
6. 您可以選擇預先設定的套件、例如 * POC 和小型工作負載 * 、 * 資料庫和應用程式資料生產工作負載 * 、 * 具成本效益的 DR* 或 * 最高效能的正式作業工作負載 * 。在此驗證中、我們選擇 * POC 和小型工作負載 * 。
7. 建立具有特定大小、允許的通訊協定和匯出選項的 Volume 。在此驗證中、我們建立了一個名為的 Volume `vol1`。
8. 選擇設定檔磁碟類型和分層原則。在此驗證中、我們停用了 * 儲存效率 * 和 * 通用 SSD ：動態效能 * 。
9. 最後、檢閱並建立 Cloud Volumes ONTAP 執行個體。然後等待 15 到 20 分鐘、讓 BlueXP 建立 Cloud Volumes ONTAP 工作環境。
10. 設定下列參數以啟用二元傳輸協定。ONTAP 9 支援二元傳輸協定（ NFS/S3 ）。12.1 及更新版本。
    a. 在此驗證中、我們建立了一個稱為的 SVM `svm_sagemaker_cvo_sn1` 和Volume `vol1`。
    b. 驗證 SVM 是否支援 NFS 和 S3 的傳輸協定。如果沒有、請修改 SVM 以支援它們。

```
sagemaker_cvo_sn1::> vserver show -vserver svm_sagemaker_cvo_sn1
                                    Vserver: svm_sagemaker_cvo_sn1
                               Vserver Type: data
                            Vserver Subtype: default
                               Vserver UUID: 911065dd-a8bc-11ed-bc24-
e1c0f00ad86b
                                Root Volume:
svm_sagemaker_cvo_sn1_root
                                  Aggregate: aggr1
                                 NIS Domain: -
                  Root Volume Security Style: unix
                                LDAP Client: -
                Default Volume Language Code: C.UTF-8
                             Snapshot Policy: default
                               Data Services: data-cifs, data-
flexcache,
                                              data-iscsi, data-nfs,
                                              data-nvme-tcp
                                     Comment:
                                Quota Policy: default
                List of Aggregates Assigned: aggr1
 Limit on Maximum Number of Volumes allowed: unlimited
                        Vserver Admin State: running
                  Vserver Operational State: running
   Vserver Operational State Stopped Reason: -
                          Allowed Protocols: nfs, cifs, fcp, iscsi,
ndmp, s3
                       Disallowed Protocols: nvme
            Is Vserver with Infinite Volume: false
                           QoS Policy Group: -
                        Caching Policy Name: -
                                Config Lock: false
                                IPspace Name: Default
                         Foreground Process: -
                    Logical Space Reporting: true
                  Logical Space Enforcement: false
Default Anti_ransomware State of the Vserver's Volumes: disabled
            Enable Analytics on New Volumes: false
    Enable Activity Tracking on New Volumes: false

sagemaker_cvo_sn1::>
```

11. 必要時建立並安裝 CA 憑證。
12. 建立服務資料原則。

```
sagemaker_cvo_sn1::*> network interface service-policy create -vserver
svm_sagemaker_cvo_sn1 -policy sagemaker_s3_nfs_policy -services data-
core,data-s3-server,data-nfs,data-flexcache
sagemaker_cvo_sn1::*> network interface create -vserver
svm_sagemaker_cvo_sn1 -lif svm_sagemaker_cvo_sn1_s3_lif -service-policy
sagemaker_s3_nfs_policy -home-node sagemaker_cvo_sn1-01 -address
172.30.10.41 -netmask 255.255.255.192

Warning: The configured failover-group has no valid failover targets for
the LIF's failover-policy. To view the failover targets for a LIF, use
         the "network interface show -failover" command.

sagemaker_cvo_sn1::*>
sagemaker_cvo_sn1::*> network interface show
Logical    Status     Network            Current       Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
sagemaker_cvo_sn1
            cluster-mgmt up/up    172.30.10.40/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            intercluster up/up    172.30.10.48/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            sagemaker_cvo_sn1-01_mgmt1
                         up/up    172.30.10.58/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
svm_sagemaker_cvo_sn1
            svm_sagemaker_cvo_sn1_data_lif
                         up/up    172.30.10.23/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            svm_sagemaker_cvo_sn1_mgmt_lif
                         up/up    172.30.10.32/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            svm_sagemaker_cvo_sn1_s3_lif
                         up/up    172.30.10.41/26    sagemaker_cvo_sn1-
```

```
01
                                                                    e0a
true
6 entries were displayed.

sagemaker_cvo_sn1::*>
sagemaker_cvo_sn1::*> vserver object-store-server create -vserver
svm_sagemaker_cvo_sn1  -is-http-enabled true -object-store-server
svm_sagemaker_cvo_s3_sn1 -is-https-enabled false
sagemaker_cvo_sn1::*> vserver object-store-server show

Vserver: svm_sagemaker_cvo_sn1

           Object Store Server Name: svm_sagemaker_cvo_s3_sn1
               Administrative State: up
                        HTTP Enabled: true
             Listener Port For HTTP: 80
                       HTTPS Enabled: false
     Secure Listener Port For HTTPS: 443
  Certificate for HTTPS Connections: -
                  Default UNIX User: pcuser
               Default Windows User: -
                            Comment:

sagemaker_cvo_sn1::*>
```

13. 檢查 Aggregate 詳細資料。

```
sagemaker_cvo_sn1::*> aggr show


Aggregate     Size Available Used% State   #Vols  Nodes            RAID
Status
--------- -------- --------- ----- ------- ------ ----------------
------------
aggr0_sagemaker_cvo_sn1_01
           124.0GB   50.88GB   59% online       1 sagemaker_cvo_
raid0,
                                                   sn1-01
normal
aggr1      907.1GB   904.9GB    0% online       2 sagemaker_cvo_
raid0,
                                                   sn1-01
normal
2 entries were displayed.

sagemaker_cvo_sn1::*>
```

14. 建立使用者和群組。

```
sagemaker_cvo_sn1::*> vserver object-store-server user create -vserver
svm_sagemaker_cvo_sn1 -user s3user

sagemaker_cvo_sn1::*> vserver object-store-server user show
Vserver     User            ID        Access Key          Secret Key
----------- --------------- --------- -------------------
-------------------
svm_sagemaker_cvo_sn1
            root            0         -                   -
   Comment: Root User
svm_sagemaker_cvo_sn1
            s3user          1         0ZNAX21JW5Q8AP80CQ2E

PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr
2 entries were displayed.

sagemaker_cvo_sn1::*>


sagemaker_cvo_sn1::*> vserver object-store-server group create -name
s3group -users s3user -comment ""

sagemaker_cvo_sn1::*>
sagemaker_cvo_sn1::*> vserver object-store-server group delete -gid 1
-vserver svm_sagemaker_cvo_sn1

sagemaker_cvo_sn1::*> vserver object-store-server group create -name
s3group -users s3user -comment "" -policies FullAccess

sagemaker_cvo_sn1::*>
```

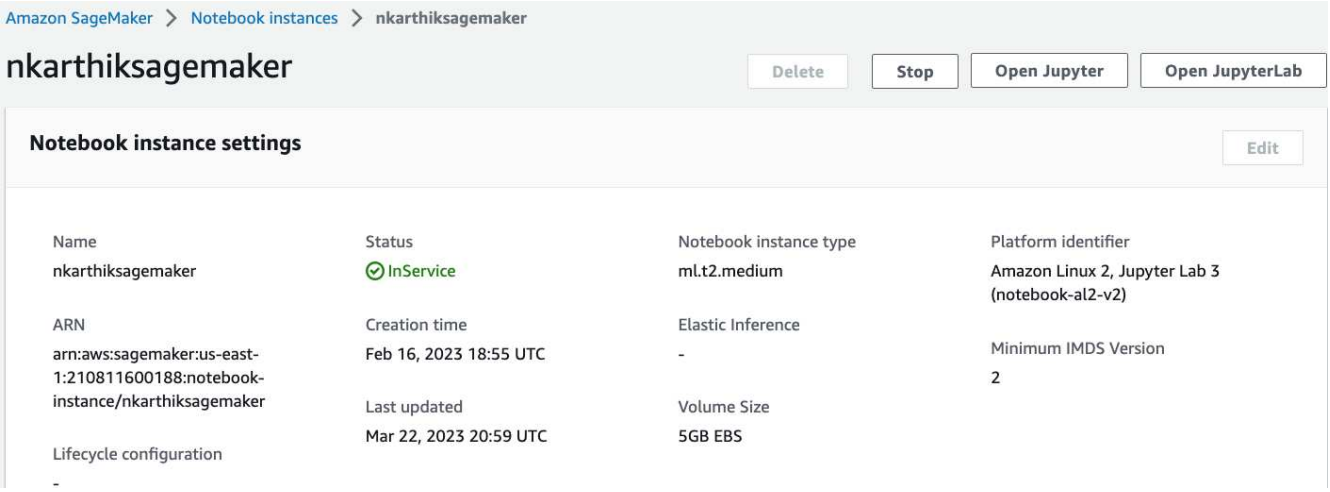
15. 在 NFS 磁碟區上建立貯體。

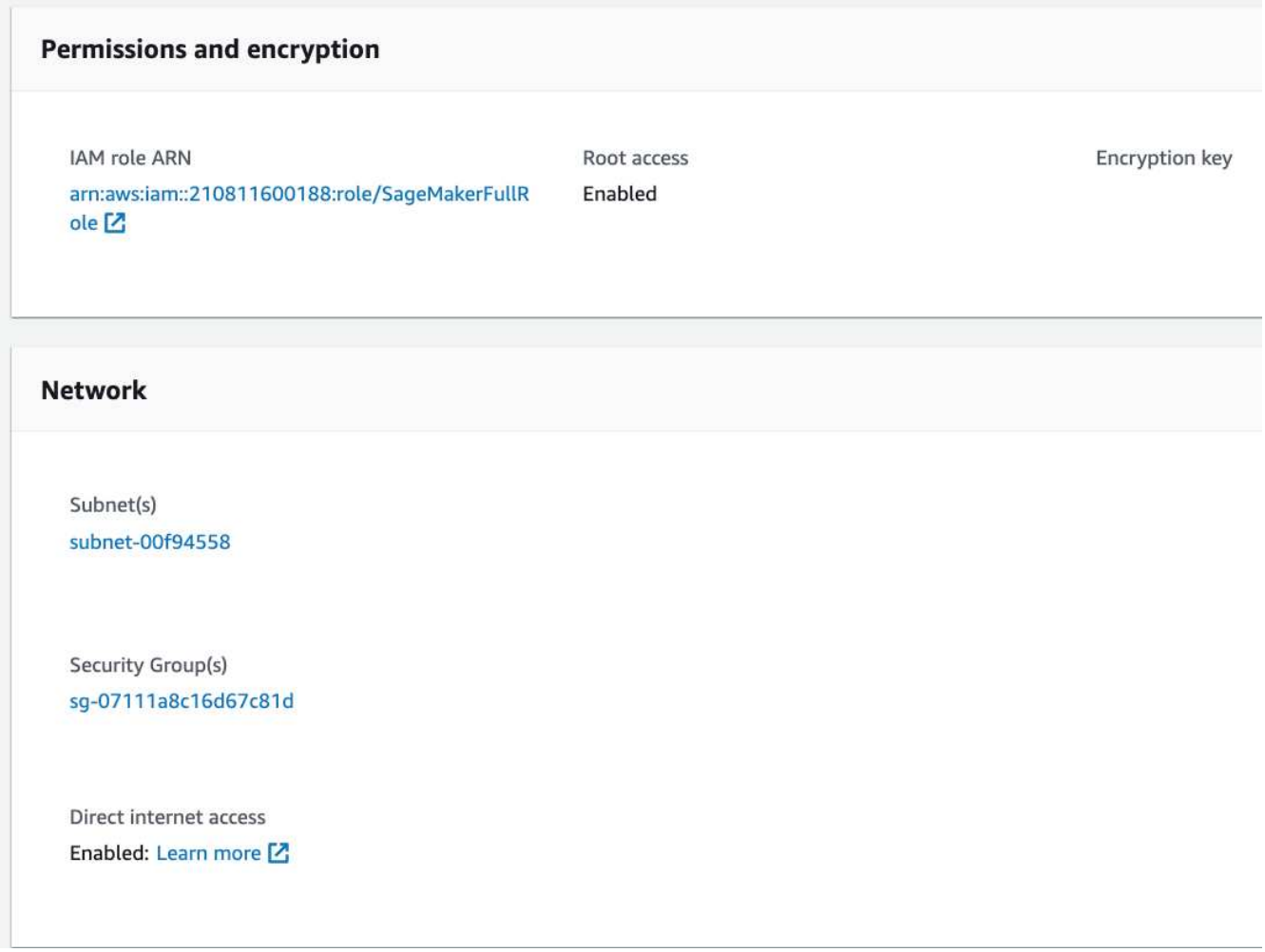
```
sagemaker_cvo_sn1::*> vserver object-store-server bucket create -bucket
ontapbucket1 -type nas -comment "" -vserver svm_sagemaker_cvo_sn1 -nas
-path /vol1
sagemaker_cvo_sn1::*> vserver object-store-server bucket show
Vserver     Bucket          Type     Volume            Size
Encryption Role       NAS Path
----------- --------------- -------- ----------------- ----------
---------- ---------- ----------
svm_sagemaker_cvo_sn1
            ontapbucket1    nas      vol1              -          false
-          /vol1
sagemaker_cvo_sn1::*>
```

AWS SageMaker

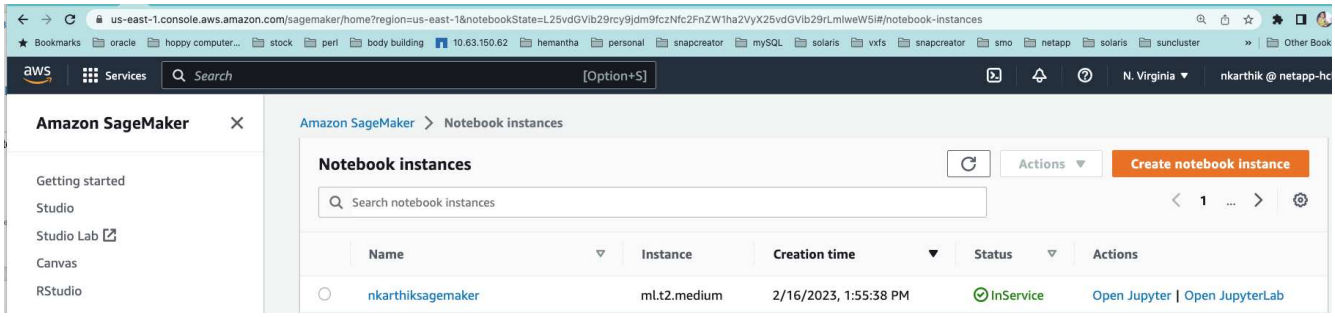
若要從 AWS SageMaker 建立 AWS 筆記型電腦、請完成下列步驟：

1. 請確定正在建立 Notebook 執行個體的使用者擁有 amzonSageMakerFullAccess IAM 原則、或是現有群組的一部分、該群組擁有 amzonSageMakerFullAccess 權限。在此驗證中、使用者是現有群組的一部分。
2. 提供下列資訊：
   - 筆記本執行個體名稱。
   - 執行個體類型。
   - 平台識別碼。
   - 選取具有 amaronSageMakerFullAccess 權限的 IAM 角色。
   - root 存取權–啟用。
   - 加密金鑰 - 選取「無自訂加密」。
   - 保留其餘的預設選項。
3. 在此驗證中、 SageMaker 執行個體詳細資料如下：

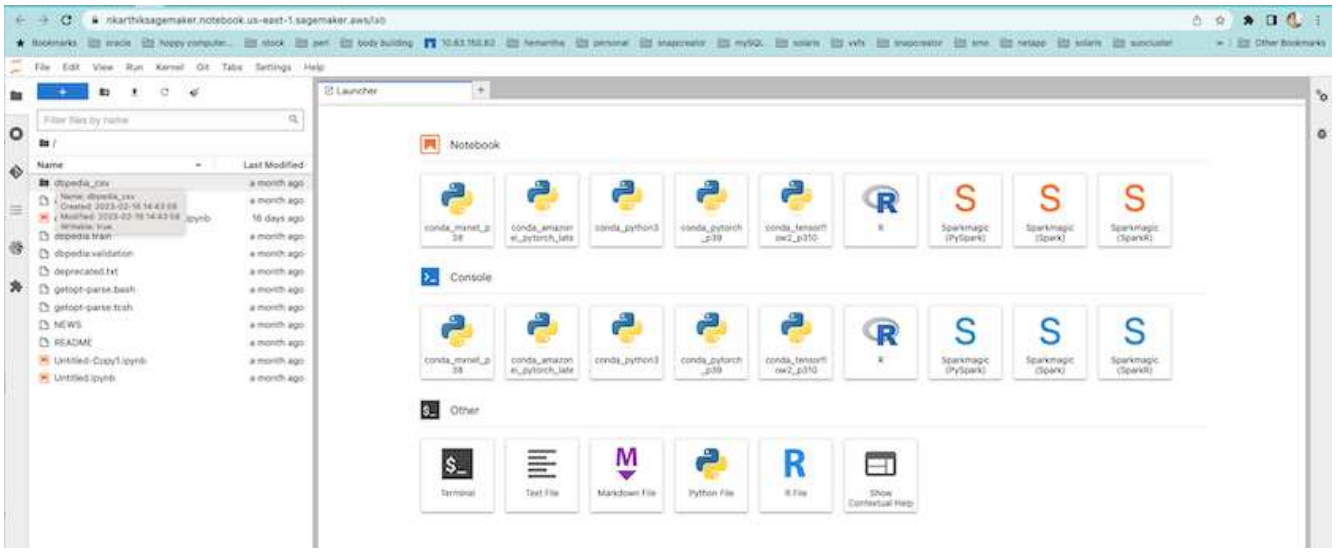




4. 啟動 AWS 筆記型電腦。



5. 開啟 Jupyter 實驗室。



6. 登入終端機並掛載 Cloud Volumes ONTAP Volume 。

```
sh-4.2$ sudo mkdir /vol1; sudo mount -t nfs 172.30.10.41:/vol1 /vol1
sh-4.2$ df -h
Filesystem          Size  Used Avail Use% Mounted on
devtmpfs            2.0G     0  2.0G   0% /dev
tmpfs               2.0G     0  2.0G   0% /dev/shm
tmpfs               2.0G  624K  2.0G   1% /run
tmpfs               2.0G     0  2.0G   0% /sys/fs/cgroup
/dev/xvda1          140G  114G   27G  82% /
/dev/xvdf           4.8G   72K  4.6G   1% /home/ec2-user/SageMaker
tmpfs               393M     0  393M   0% /run/user/1001
tmpfs               393M     0  393M   0% /run/user/1002
tmpfs               393M     0  393M   0% /run/user/1000
172.30.10.41:/vol1  973M  189M  785M  20% /vol1
sh-4.2$
```

7. 使用 AWS CLI 命令檢查在 Cloud Volumes ONTAP 磁碟區上建立的貯體。

```
sh-4.2$ aws configure --profile netapp
AWS Access Key ID [None]: 0ZNAX21JW5Q8AP80CQ2E
AWS Secret Access Key [None]: PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr
Default region name [None]: us-east-1
Default output format [None]:
sh-4.2$

sh-4.2$ aws s3 ls --profile netapp --endpoint-url
2023-02-10 17:59:48 ontapbucket1

sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/


2023-02-10 18:46:44       4747 1
2023-02-10 18:48:32         96 setup.cfg

sh-4.2$
```

用於機器學習的資料

在這項驗證中、我們使用來自 DBexpedia 的資料集、這是一項來自群眾的社群努力、從各種 Wikimedia 專案所建立的資訊中擷取結構化內容。

1. 從 DBexpedia GitHub 位置下載資料並將其解壓縮。請使用上一節所使用的相同終端機。

```
sh-4.2$ wget
--2023-02-14 23:12:11--
Resolving github.com (github.com)... 140.82.113.3
Connecting to github.com (github.com)|140.82.113.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location:  [following]
--2023-02-14 23:12:11--
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 68431223 (65M) [application/octet-stream]
Saving to: ‘dbpedia_csv.tar.gz’

100%[===================================================================
===================================================================
===================>] 68,431,223  56.2MB/s   in 1.2s

2023-02-14 23:12:13 (56.2 MB/s) - ‘dbpedia_csv.tar.gz’ saved
[68431223/68431223]

sh-4.2$ tar -zxvf dbpedia_csv.tar.gz
dbpedia_csv/
dbpedia_csv/test.csv
dbpedia_csv/classes.txt
dbpedia_csv/train.csv
dbpedia_csv/readme.txt
sh-4.2$
```

2. 將資料複製到 Cloud Volumes ONTAP 位置、然後使用 AWS CLI 從 S3 儲存區檢查資料。

```
sh-4.2$ df -h
Filesystem          Size  Used Avail Use% Mounted on
devtmpfs            2.0G     0  2.0G   0% /dev
tmpfs               2.0G     0  2.0G   0% /dev/shm
tmpfs               2.0G  628K  2.0G   1% /run
tmpfs               2.0G     0  2.0G   0% /sys/fs/cgroup
/dev/xvda1          140G  114G   27G  82% /
/dev/xvdf           4.8G   52K  4.6G   1% /home/ec2-user/SageMaker
tmpfs               393M     0  393M   0% /run/user/1002
tmpfs               393M     0  393M   0% /run/user/1001
tmpfs               393M     0  393M   0% /run/user/1000
172.30.10.41:/vol1  973M  384K  973M   1% /vol1
sh-4.2$ pwd
/home/ec2-user
sh-4.2$ cp -ra dbpedia_csv /vol1
sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/
                           PRE dbpedia_csv/
2023-02-10 18:46:44       4747 1
2023-02-10 18:48:32         96 setup.cfg
sh-4.2$
```

3. 執行基本驗證、確保 S3 儲存區的讀取 / 寫入功能正常運作。

```
sh-4.2$ aws s3 cp  --profile netapp --endpoint-url  /usr/share/doc/util-
linux-2.30.2 s3://ontapbucket1/ --recursive
upload: ../../../usr/share/doc/util-linux-2.30.2/deprecated.txt to
s3://ontapbucket1/deprecated.txt
upload: ../../../usr/share/doc/util-linux-2.30.2/getopt-parse.bash to
s3://ontapbucket1/getopt-parse.bash
upload: ../../../usr/share/doc/util-linux-2.30.2/README to
s3://ontapbucket1/README
upload: ../../../usr/share/doc/util-linux-2.30.2/getopt-parse.tcsh to
s3://ontapbucket1/getopt-parse.tcsh
upload: ../../../usr/share/doc/util-linux-2.30.2/AUTHORS to
s3://ontapbucket1/AUTHORS
upload: ../../../usr/share/doc/util-linux-2.30.2/NEWS to
s3://ontapbucket1/NEWS
sh-4.2$ aws s3 ls --profile netapp --endpoint-url
s3://ontapbucket1/s3://ontapbucket1/

An error occurred (InternalError) when calling the ListObjectsV2
operation: We encountered an internal error. Please try again.
sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/
                           PRE dbpedia_csv/
```

```
2023-02-16 19:19:27      26774 AUTHORS
2023-02-16 19:19:27      72727 NEWS
2023-02-16 19:19:27       4493 README
2023-02-16 19:19:27       2825 deprecated.txt
2023-02-16 19:19:27       1590 getopt-parse.bash
2023-02-16 19:19:27       2245 getopt-parse.tcsh
sh-4.2$ ls -ltr /vol1
total 132
drwxrwxr-x 2 ec2-user ec2-user  4096 Mar 29  2015 dbpedia_csv
-rw-r--r-- 1 nobody   nobody    2245 Apr 10 17:37 getopt-parse.tcsh
-rw-r--r-- 1 nobody   nobody    2825 Apr 10 17:37 deprecated.txt
-rw-r--r-- 1 nobody   nobody    4493 Apr 10 17:37 README
-rw-r--r-- 1 nobody   nobody    1590 Apr 10 17:37 getopt-parse.bash
-rw-r--r-- 1 nobody   nobody   26774 Apr 10 17:37 AUTHORS
-rw-r--r-- 1 nobody   nobody   72727 Apr 10 17:37 NEWS
sh-4.2$ ls -ltr /vol1/dbpedia_csv/
total 192104
-rw------- 1 ec2-user ec2-user 174148970 Mar 28  2015 train.csv
-rw------- 1 ec2-user ec2-user  21775285 Mar 28  2015 test.csv
-rw------- 1 ec2-user ec2-user       146 Mar 28  2015 classes.txt
-rw-rw-r-- 1 ec2-user ec2-user      1758 Mar 29  2015 readme.txt
sh-4.2$ chmod -R 777 /vol1/dbpedia_csv
sh-4.2$ ls -ltr /vol1/dbpedia_csv/
total 192104
-rwxrwxrwx 1 ec2-user ec2-user 174148970 Mar 28  2015 train.csv
-rwxrwxrwx 1 ec2-user ec2-user  21775285 Mar 28  2015 test.csv
-rwxrwxrwx 1 ec2-user ec2-user       146 Mar 28  2015 classes.txt
-rwxrwxrwx 1 ec2-user ec2-user      1758 Mar 29  2015 readme.txt
sh-4.2$ aws s3 cp --profile netapp --endpoint-url http://172.30.2.248/
s3://ontapbucket1/ /tmp --recursive
download: s3://ontapbucket1/AUTHORS to ../../tmp/AUTHORS
download: s3://ontapbucket1/README to ../../tmp/README
download: s3://ontapbucket1/NEWS to ../../tmp/NEWS
download: s3://ontapbucket1/dbpedia_csv/classes.txt to
../../tmp/dbpedia_csv/classes.txt
download: s3://ontapbucket1/dbpedia_csv/readme.txt to
../../tmp/dbpedia_csv/readme.txt
download: s3://ontapbucket1/deprecated.txt to ../../tmp/deprecated.txt
download: s3://ontapbucket1/getopt-parse.bash to ../../tmp/getopt-
parse.bash
download: s3://ontapbucket1/getopt-parse.tcsh to ../../tmp/getopt-
parse.tcsh
download: s3://ontapbucket1/dbpedia_csv/test.csv to
../../tmp/dbpedia_csv/test.csv
download: s3://ontapbucket1/dbpedia_csv/train.csv to
../../tmp/dbpedia_csv/train.csv
```

```
sh-4.2$
sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/
                           PRE dbpedia_csv/
2023-02-16 19:19:27      26774 AUTHORS
2023-02-16 19:19:27      72727 NEWS
2023-02-16 19:19:27       4493 README
2023-02-16 19:19:27       2825 deprecated.txt
2023-02-16 19:19:27       1590 getopt-parse.bash
2023-02-16 19:19:27       2245 getopt-parse.tcsh
sh-4.2$
```

驗證 Jupyter 筆記型電腦的機器學習

下列驗證功能可透過以下 SageMaker BlazingText 範例、透過文字分類提供機器學習建置、訓練及部署模型：

1. 安裝 boto3 和 SageMaker 套件。

```
In [1]:  pip install --upgrade boto3 sagemaker
```

輸出：

```
Looking in indexes: https://pypi.org/simple,
https://pip.repos.neuron.amazo naws.com
Requirement already satisfied: boto3 in /home/ec2-
user/anaconda3/envs/pytho n3/lib/python3.10/site-packages (1.26.44)
Collecting boto3
  Downloading boto3-1.26.72-py3-none-any.whl (132 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
132.7/132.7 kB 14.6 MB/s eta 0: 00:00
Requirement already satisfied: sagemaker in /home/ec2-
user/anaconda3/envs/p ython3/lib/python3.10/site-packages (2.127.0)
Collecting sagemaker
  Downloading sagemaker-2.132.0.tar.gz (668 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
668.0/668.0 kB 12.3 MB/s eta 0:
00:0000:01
  Preparing metadata (setup.py) ... done
Collecting botocore<1.30.0,>=1.29.72
  Downloading botocore-1.29.72-py3-none-any.whl (10.4 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
10.4/10.4 MB 44.3 MB/s eta 0: 00:0000:010:01
Requirement already satisfied: s3transfer<0.7.0,>=0.6.0 in /home/ec2-
user/a naconda3/envs/python3/lib/python3.10/site-packages (from boto3)
(0.6.0)
```

```
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /home/ec2-
user/ana conda3/envs/python3/lib/python3.10/site-packages (from boto3)
(0.10.0)
Requirement already satisfied: attrs<23,>=20.3.0 in /home/ec2-
user/anaconda
3/envs/python3/lib/python3.10/site-packages (from sagemaker) (22.1.0)
Requirement already satisfied: google-pasta in /home/ec2-
user/anaconda3/env s/python3/lib/python3.10/site-packages (from
sagemaker) (0.2.0)
Requirement already satisfied: numpy<2.0,>=1.9.0 in /home/ec2-
user/anaconda
3/envs/python3/lib/python3.10/site-packages (from sagemaker) (1.22.4)
Requirement already satisfied: protobuf<4.0,>=3.1 in /home/ec2-
user/anacond a3/envs/python3/lib/python3.10/site-packages (from
sagemaker) (3.20.3)
Requirement already satisfied: protobuf3-to-dict<1.0,>=0.1.5 in
/home/ec2-u ser/anaconda3/envs/python3/lib/python3.10/site-packages
(from sagemaker)
(0.1.5)
Requirement already satisfied: smdebug_rulesconfig==1.0.1 in /home/ec2-
use r/anaconda3/envs/python3/lib/python3.10/site-packages (from
sagemaker) (1.
0.1) Requirement already satisfied: importlib-metadata<5.0,>=1.4.0 in
/home/ec2user/anaconda3/envs/python3/lib/python3.10/site-packages (from
sagemaker)
(4.13.0)
Requirement already satisfied: packaging>=20.0 in /home/ec2-
user/anaconda3/ envs/python3/lib/python3.10/site-packages (from
sagemaker) (21.3)
Requirement already satisfied: pandas in /home/ec2-
user/anaconda3/envs/pyth on3/lib/python3.10/site-packages (from
sagemaker) (1.5.1)
Requirement already satisfied: pathos in /home/ec2-
user/anaconda3/envs/pyth on3/lib/python3.10/site-packages (from
sagemaker) (0.3.0)
Requirement already satisfied: schema in /home/ec2-
user/anaconda3/envs/pyth on3/lib/python3.10/site-packages (from
sagemaker) (0.7.5) Requirement already satisfied: python-
dateutil<3.0.0,>=2.1 in /home/ec2-use
r/anaconda3/envs/python3/lib/python3.10/site-packages (from
botocore<1.30.
0,>=1.29.72->boto3) (2.8.2)
Requirement already satisfied: urllib3<1.27,>=1.25.4 in /home/ec2-
user/anac onda3/envs/python3/lib/python3.10/site-packages (from
botocore<1.30.0,>=1.2
9.72->boto3) (1.26.8) Requirement already satisfied: zipp>=0.5 in
```

```
/home/ec2-user/anaconda3/envs/p ython3/lib/python3.10/site-packages
(from importlib-metadata<5.0,>=1.4.0->s agemaker) (3.10.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in /home/ec2-
user/a naconda3/envs/python3/lib/python3.10/site-packages (from
packaging>=20.0->s agemaker) (3.0.9)
Requirement already satisfied: six in /home/ec2-
user/anaconda3/envs/python
3/lib/python3.10/site-packages (from protobuf3-to-dict<1.0,>=0.1.5-
>sagemak er) (1.16.0)
Requirement already satisfied: pytz>=2020.1 in /home/ec2-
user/anaconda3/env s/python3/lib/python3.10/site-packages (from pandas-
>sagemaker) (2022.5)
Requirement already satisfied: ppft>=1.7.6.6 in /home/ec2-
user/anaconda3/en vs/python3/lib/python3.10/site-packages (from pathos-
>sagemaker) (1.7.6.6) Requirement already satisfied:
multiprocess>=0.70.14 in /home/ec2-user/anac
onda3/envs/python3/lib/python3.10/site-packages (from pathos->sagemaker)
(0.70.14)
Requirement already satisfied: dill>=0.3.6 in /home/ec2-
user/anaconda3/env s/python3/lib/python3.10/site-packages (from pathos-
>sagemaker) (0.3.6)
Requirement already satisfied: pox>=0.3.2 in /home/ec2-
user/anaconda3/envs/ python3/lib/python3.10/site-packages (from pathos-
>sagemaker) (0.3.2) Requirement already satisfied: contextlib2>=0.5.5 in
/home/ec2-user/anacond a3/envs/python3/lib/python3.10/site-packages
(from schema->sagemaker) (21.
6.0) Building wheels for collected packages: sagemaker
  Building wheel for sagemaker (setup.py) ... done
  Created wheel for sagemaker: filename=sagemaker-2.132.0-py2.py3-none-
any. whl size=905449
sha256=f6100a5dc95627f2e2a49824e38f0481459a27805ee19b5a06ec
83db0252fd41
  Stored in directory: /home/ec2-
user/.cache/pip/wheels/60/41/b6/482e7ab096
520df034fbf2dddd244a1d7ba0681b27ef45aa61
Successfully built sagemaker
Installing collected packages: botocore, boto3, sagemaker
  Attempting uninstall: botocore     Found existing installation:
botocore 1.24.19
    Uninstalling botocore-1.24.19:       Successfully uninstalled
botocore-1.24.19
  Attempting uninstall: boto3     Found existing installation: boto3
1.26.44
    Uninstalling boto3-1.26.44:
      Successfully uninstalled boto3-1.26.44
  Attempting uninstall: sagemaker     Found existing installation:
```

```
sagemaker 2.127.0
    Uninstalling sagemaker-2.127.0:
      Successfully uninstalled sagemaker-2.127.0
ERROR: pip's dependency resolver does not currently take into account
all t he packages that are installed. This behaviour is the source of
the followi ng dependency conflicts.
awscli 1.27.44 requires botocore==1.29.44, but you have botocore 1.29.72
wh ich is incompatible.
aiobotocore 2.0.1 requires botocore<1.22.9,>=1.22.8, but you have
botocore 1.29.72 which is incompatible. Successfully installed boto3-
1.26.72 botocore-1.29.72 sagemaker-2.132.0 Note: you may need to restart
the kernel to use updated packages.
```

2. 在下列步驟中、資料 (`dbpedia_csv`）從 S3 儲存區下載 `ontapbucket1` 至用於機器學習的 Jupyter Notebook 執行個體。

```
In [2]: import sagemaker
In [3]: from sagemaker import get_execution_role
In [4]:
import json
import boto3
sess = sagemaker.Session()
role = get_execution_role()
print(role)
bucket = "ontapbucket1"
print(bucket)
sess.s3_client = boto3.client('s3',region_name='',aws_access_key_id =
'0ZNAX21JW5Q8AP80CQ2E',  aws_secret_access_key =
'PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr',
                              use_ssl = False, endpoint_url =
'http://172.30.10.41',

config=boto3.session.Config(signature_version='s3v4',
s3={'addressing_style':'path'}) )
sess.s3_resource = boto3.resource('s3',region_name='',aws_access_key_id
= '0ZNAX21JW5Q8AP80CQ2E', aws_secret_access_key =
'PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr',
                              use_ssl = False, endpoint_url =
'http://172.30.10.41',

config=boto3.session.Config(signature_version='s3v4',
s3={'addressing_style':'path'}) )
prefix = "blazingtext/supervised"
import os
my_bucket = sess.s3_resource.Bucket(bucket)
my_bucket = sess.s3_resource.Bucket(bucket)
#os.mkdir('dbpedia_csv')
for s3_object in my_bucket.objects.all():
    filename = s3_object.key
#    print(filename)
#    print(s3_object.key)
    my_bucket.download_file(s3_object.key, filename)
```

3. 下列程式碼會建立從整數索引到類別標籤的對應、以便在推斷期間擷取實際類別名稱。

```
index_to_label = {}
with open("dbpedia_csv/classes.txt") as f:
    for i,label in enumerate(f.readlines()):
        index_to_label[str(i + 1)] = label.strip()
```

輸出會列出中的檔案和資料夾 `ontapbucket1` 做為 AWS SageMaker 機器學習驗證資料的貯體。

```
arn:aws:iam::210811600188:role/SageMakerFullRole ontapbucket1
AUTHORS
AUTHORS
NEWS
NEWS
README README
dbpedia_csv/classes.txt dbpedia_csv/classes.txt dbpedia_csv/readme.txt
dbpedia_csv/readme.txt dbpedia_csv/test.csv dbpedia_csv/test.csv
dbpedia_csv/train.csv dbpedia_csv/train.csv deprecated.txt
deprecated.txt getopt-parse.bash getopt-parse.bash getopt-parse.tcsh
getopt-parse.tcsh
In [5]: ls
AUTHORS       deprecated.txt     getopt-parse.tcsh  NEWS
Untitled.ipynb dbpedia_csv/  getopt-parse.bash  lost+found/
README
In [6]: ls -l dbpedia_csv
total 191344
-rw-rw-r-- 1 ec2-user ec2-user       146 Feb 16 19:43 classes.txt
-rw-rw-r-- 1 ec2-user ec2-user      1758 Feb 16 19:43 readme.txt
-rw-rw-r-- 1 ec2-user ec2-user  21775285 Feb 16 19:43 test.csv
-rw-rw-r-- 1 ec2-user ec2-user 174148970 Feb 16 19:43 train.csv
```

4. 開始資料預先處理階段、將訓練資料預先處理成空間分隔、可由 BlazingText 演算法和 nltk 程式庫使用的權證化文字格式、以使 DBPedia 資料集的輸入句子變成權證。下載 nltk tokenizer 和其他程式庫。。`transform_instance` 平行套用至每個資料執行個體使用 Python 多重處理模組。

```
ln [7]: from random import shuffle
import multiprocessing
from multiprocessing import Pool
import csv
import nltk
nltk.download("punkt")
def transform_instance(row):
    cur_row = []
    label ="__label__" + index_to_label [row[0]] # Prefix the index-ed
label with __label__
    cur_row.append (label)
    cur_row.extend(nltk.word_tokenize(row[1].lower ()))
    cur_row.extend(nltk.word_tokenize(row[2].lower ()))
    return cur_row
def preprocess(input_file, output_file, keep=1):
    all_rows = []
    with open(input_file,"r") as csvinfile:
```

```
        csv_reader = csv.reader(csvinfile, delimiter=",")
        for row in csv_reader:
            all_rows.append(row)
    shuffle(all_rows)
    all_rows = all_rows[: int(keep * len(all_rows))]
    pool = Pool(processes=multiprocessing.cpu_count())
    transformed_rows = pool.map(transform_instance, all_rows)
    pool.close()
    pool. join()
    with open(output_file, "w") as csvoutfile:
        csv_writer = csv.writer (csvoutfile, delimiter=" ",
lineterminator="\n")
        csv_writer.writerows (transformed_rows)

# Preparing the training dataset
# since preprocessing the whole dataset might take a couple of minutes,
# we keep 20% of the training dataset for this demo.
# Set keep to 1 if you want to use the complete dataset
preprocess("dbpedia_csv/train.csv","dbpedia.train", keep=0.2)
# Preparing the validation dataset
preprocess("dbpedia_csv/test.csv","dbpedia.validation")
sess = sagemaker.Session()
role = get_execution_role()
print (role) # This is the role that sageMaker would use to leverage Aws
resources (S3,  Cloudwatch) on your behalf
bucket = sess.default_bucket() # Replace with your own bucket name if
needed
print("default Bucket::: ")
print(bucket)
```

輸出：

```
[nltk_data] Downloading package punkt to /home/ec2-user/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
arn:aws:iam::210811600188:role/SageMakerFullRole default Bucket:::
sagemaker-us-east-1-210811600188
```

5. 將格式化和訓練資料集上傳至 S3 、讓 SageMaker 可以使用該資料集來執行訓練工作。然後使用 Python SDK 將兩個檔案上傳至貯體和前置碼位置。

```
ln [8]: %%time
train_channel = prefix + "/train"
validation_channel = prefix + "/validation"
sess.upload_data(path="dbpedia.train", bucket=bucket,
key_prefix=train_channel)
sess.upload_data(path="dbpedia.validation", bucket=bucket,
key_prefix=validation_channel)
s3_train_data = "s3://{}/{}".format(bucket, train_channel)
s3_validation_data = "s3://{}/{}".format(bucket, validation_channel)
```

輸出：

```
CPU times: user 546 ms, sys: 163 ms, total: 709 ms
Wall time: 1.32 s
```

6. 在 S3 上設定輸出位置、將模型成品載入其中、使成品能成為演算法訓練工作的輸出。建立 `sageMaker.estimator.Estimator` 物件以啟動訓練工作。

```
In [9]: s3_output_location = "s3://{}/{}/output".format(bucket, prefix)
In [10]: region_name = boto3.Session().region_name
In [11]: container =
sagemaker.amazon.amazon_estimator.get_image_uri(region_name,
"blazingtext","latest")
print("Using SageMaker BlazingText container: {} ({})".format(container,
region_name))
```

輸出：

```
The method get_image_uri has been renamed in sagemaker>=2.
See: https://sagemaker.readthedocs.io/en/stable/v2.html for details.
Defaulting to the only supported framework/algorithm version: 1.
Ignoring f ramework/algorithm version: latest.
Using SageMaker BlazingText container: 811284229777.dkr.ecr.us-east-
1.amazo naws.com/blazingtext:1 (us-east-1)
```

7. 定義 SageMaker `Estrimator` 使用資源組態和超參數、在 c4.4xlarge 執行個體上使用受監督模式、在 DBPedia 資料集上訓練文字分類。

```
In [12]: bt_model = sagemaker.estimator.Estimator(
container,
role,
instance_count=1,
instance_type="ml.c4.4xlarge",
volume_size=30,
max_run=360000,
input_mode="File",
output_path=s3_output_location,
hyperparameters={
        "mode": "supervised",
        "epochs": 1,
        "min_count": 2,
        "learning_rate": 0.05,
        "vector_dim": 10,
        "early_stopping": True,
        "patience": 4,
        "min_epochs": 5,
        "word_ngrams": 2,
 },
     )
```

8. 準備資料通道與演算法之間的交握。若要這麼做、請建立 `sagemaker.session.s3_input` 來自資料通道的物件、並將其保留在字典中、以供演算法使用。

```
ln [13]: train_data = sagemaker.inputs.TrainingInput(
    s3_train_data,
    distribution="FullyReplicated",
    content_type="text/plain",
    s3_data_type="S3Prefix",
)
validation_data = sagemaker.inputs.TrainingInput(
    s3_validation_data,
    distribution="FullyReplicated",
    content_type="text/plain",
    s3_data_type="S3Prefix",
)
data_channels = {"train": train_data, "validation": validation_data}
```

9. 工作完成後、會出現「工作完成」訊息。您可以在設定為的 S3 儲存貯體中找到經過訓練的機型 `output_path` 在評估者中。

```
ln [14]: bt_model.fit(inputs=data_channels, logs=True)
```

輸出：

```
INFO:sagemaker:Creating training-job with name: blazingtext-2023-02-16-
20-3
7-30-748
2023-02-16 20:37:30 Starting - Starting the training job......
2023-02-16 20:38:09 Starting - Preparing the instances for
training......
2023-02-16 20:39:24 Downloading - Downloading input data
2023-02-16 20:39:24 Training - Training image download completed.
Training in progress... Arguments: train
[02/16/2023 20:39:41 WARNING 140279908747072] Loggers have already been
set up. [02/16/2023 20:39:41 WARNING 140279908747072] Loggers have
already been set up.
[02/16/2023 20:39:41 INFO 140279908747072] nvidia-smi took:
0.0251793861389
16016 secs to identify 0 gpus
[02/16/2023 20:39:41 INFO 140279908747072] Running single machine CPU
Blazi ngText training using supervised mode.
Number of CPU sockets found in instance is  1
[02/16/2023 20:39:41 INFO 140279908747072] Processing
/opt/ml/input/data/tr ain/dbpedia.train . File size: 35.0693244934082 MB
[02/16/2023 20:39:41 INFO 140279908747072] Processing
/opt/ml/input/data/va lidation/dbpedia.validation . File size:
21.887572288513184 MB
Read 6M words
Number of words:  149301
Loading validation data from
/opt/ml/input/data/validation/dbpedia.validati on
Loaded validation data.
-------------- End of epoch: 1 ##### Alpha: 0.0000  Progress: 100.00%
Million Words/sec: 10.39 ##### Training finished.
Average throughput in Million words/sec: 10.39
Total training time in seconds: 0.60
#train_accuracy: 0.7223
Number of train examples: 112000
#validation_accuracy: 0.7205
Number of validation examples: 70000
2023-02-16 20:39:55 Uploading - Uploading generated training model
2023-02-16 20:40:11 Completed - Training job completed
Training seconds: 68
Billable seconds: 68
```

10. 訓練完成後、請將經過訓練的模型部署為 Amazon SageMaker 即時代管端點、以做出預測。

```
In [15]: from sagemaker.serializers import JSONSerializer
 text_classifier = bt_model.deploy(
     initial_instance_count=1, instance_type="ml.m4.xlarge",
serializer=JSONS
)
```

輸出：

```
INFO:sagemaker:Creating model with name: blazingtext-2023-02-16-20-41-
33-10
0
INFO:sagemaker:Creating endpoint-config with name blazingtext-2023-02-
16-20
-41-33-100
INFO:sagemaker:Creating endpoint with name blazingtext-2023-02-16-20-41-
33-
100
-------!
```

```
In [16]: sentences = [
    "Convair was an american aircraft manufacturing company which later
expanded into rockets and spacecraft.",
       "Berwick secondary college is situated in the outer melbourne
metropolitan suburb of berwick .",
]
# using the same nltk tokenizer that we used during data preparation for
training
tokenized_sentences = [" ".join(nltk.word_tokenize(sent)) for sent in
sentences]
payload = {"instances": tokenized_sentences} response =
text_classifier.predict(payload)
predictions = json.loads(response)
print(json.dumps(predictions, indent=2))
```

```
[
  {
    "label": [
      "__label__Artist"
    ],
    "prob": [
      0.4090951681137085
    ]
  },
  {
    "label": [
      "__label__EducationalInstitution"
    ],
    "prob": [
      0.49466073513031006
    ]
  }
]
```

11. 根據預設，模型會傳回一個機率最高的預測值。以擷取頂端 `k` 預測、設定 `k` 在組態檔案中。

```
In [17]: payload = {"instances": tokenized_sentences, "configuration":
{"k": 2}}
 response = text_classifier.predict(payload)

 predictions = json.loads(response)
 print(json.dumps(predictions, indent=2))
```

```
[
  {
    "label": [
      "__label__Artist",
      "__label__MeanOfTransportation"
    ],
    "prob": [
      0.4090951681137085,
      0.26930734515190125
    ]
  },
  {
    "label": [
      "__label__EducationalInstitution",
      "__label__Building"
    ],
    "prob": [
      0.49466073513031006,
      0.15817692875862122
    ]
  }
]
```

12. 在關閉筆記本之前刪除端點。

```
In [18]: sess.delete_endpoint(text_classifier.endpoint)
WARNING:sagemaker.deprecations:The endpoint attribute has been renamed
in s agemaker>=2.
See: https://sagemaker.readthedocs.io/en/stable/v2.html for details.
INFO:sagemaker:Deleting endpoint with name: blazingtext-2023-02-16-20-
41-33
-100
```

結論

根據這項驗證、資料科學家和工程師可透過 NetApp Cloud Volumes ONTAP 的 S3 儲存區、從 AWS SageMaker Jupyter 筆記型電腦存取 NFS 資料。這種方法可讓您輕鬆從 NFS 和 S3 存取和共用相同的資料、而不需要額外的軟體。

何處可找到其他資訊

若要深入瞭解本文所述資訊、請檢閱下列文件和 / 或網站：

- 使用 SageMaker BlazingText 進行文字分類

  "https://sagemaker-examples.readthedocs.io/en/latest/introduction_to_amazon_algorithms/blazingtext_text_classification_dbpedia/blazingtext_text_classification_dbpedia.html"

- 支援S3物件儲存的版本ONTAP

  "https://docs.netapp.com/us-en/ontap/s3-config/ontap-version-support-s3-concept.html"

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